

✓ NUMPY

```
import numpy as np
```

```
#numpy is like an array of number  
a=np.array([1,2,3])  
print(type(a), a.shape, a[1])  
a[0]=10# reading or writing  
print(a)
```

```
<class 'numpy.ndarray'> (3,) 2  
[10  2  3]
```

```
b=np.array([[1,2,3],[4,5,6]])  
print(b)  
print(b.shape)  
#[:,:]
```

```
[[1 2 3]  
 [4 5 6]]  
(2, 3)
```

```
print(b[0,0],b[1,1])
```

```
1 5
```

```
n1=np.zeros((2,2))  
print(n1)
```

```
[[0. 0.]  
 [0. 0.]]
```

```
n2=np.ones((1,2))  
print(n2)
```

```
[[1. 1.]]
```

```
n3=np.full((2,2),7)  
print(n3)
```

```
[[7 7]  
 [7 7]]
```

```
n4=np.eye(3)
print(n4)
```

```
[[1.  0.  0.]
 [0.  1.  0.]
 [0.  0.  1.]]
```

```
num4=np.random.random((2,2))
print(num4)
```

```
[[0.25280486 0.45535266]
 [0.68978752 0.82855071]]
```

```
any_var=b[1,:2]
```

```
print(any_var)
```

```
[4 5]
```

Start coding or [generate](#) with AI.